Supplementary Materials: Banach-Space Norm Descriptors for Interferometric Profile Metrology

1. Supplementary Experimental Details

1.1. Rotation stage specifications

Table 1: Technical specifications of the rotation stage [?].

Parameter	Specification
Angular range	Unrestricted number of revolutions
Maximum speed	80°/s
Encoder accuracy	2.0 mdeg
Wobble, typical	± 12 μrad
Wobble, guaranteed	± 25 μrad
Eccentricity, typical	± 0.40 μm
Transversal compliance	10 μrad/N m
Origin repeatability	± 0.25 mdeg

1.2. Environmental shielding and vibration isolation

Figure 1: Setup shielded by a plexiglass box and covered by a wool blanket.

Figure 2: Layered wooden-polystyrene support with granite plate.

1.3. Single-image radial phase extraction

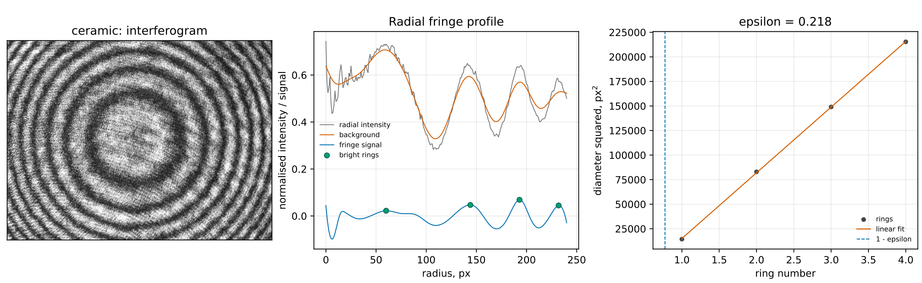

Figure 3: Ceramic single-image radial phase extraction. (Left) Raw interferogram (648×488 px). (Centre) Radial fringe intensity after background subtraction and B-spline smoothing (blue curve); red dots mark detected bright-ring maxima. (Right) D_j^2–order linear fit (blue line, red points): the $D^2 = 0$ intercept (dashed) gives $1 - \epsilon$.

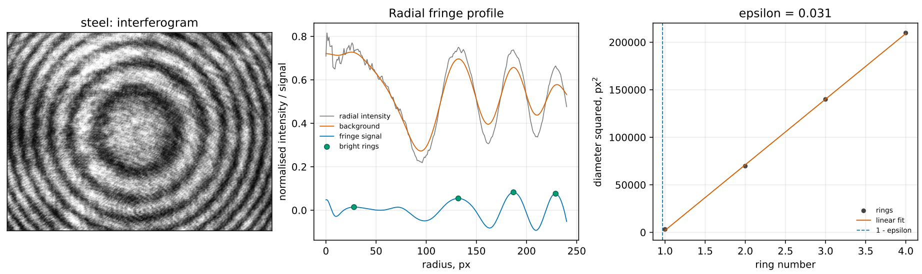

Figure 4: Steel single-image radial phase extraction. Same EFM procedure applied to a steel interferogram (640×480 px).

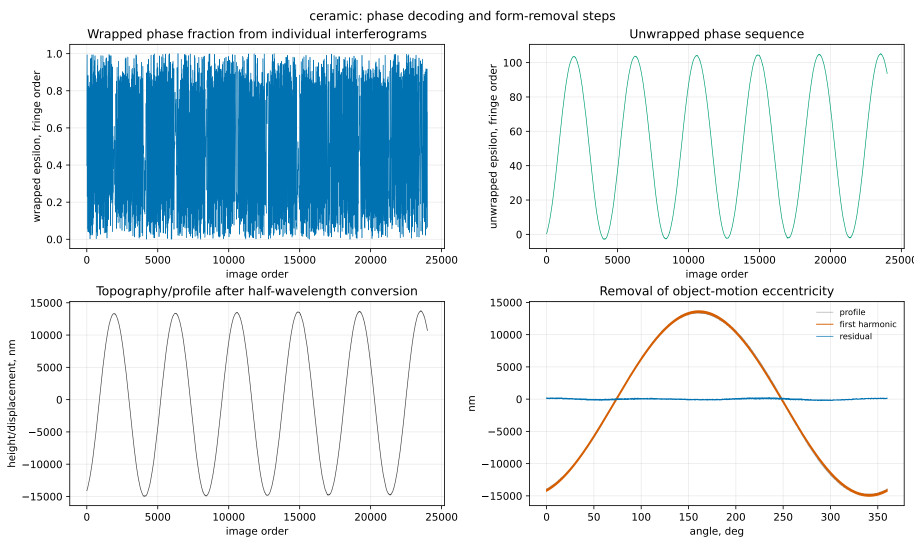

Figure 5: Ceramic sequence processing. (Top left) Wrapped excess fraction. (Top right) Unwrapped fringe order. (Bottom left) Nanometre height profile with LSCI reference circle and eccentricity fit. (Bottom right) Gaussian low-pass filtered residual (50% at 15 UPR, ISO 16610-21).

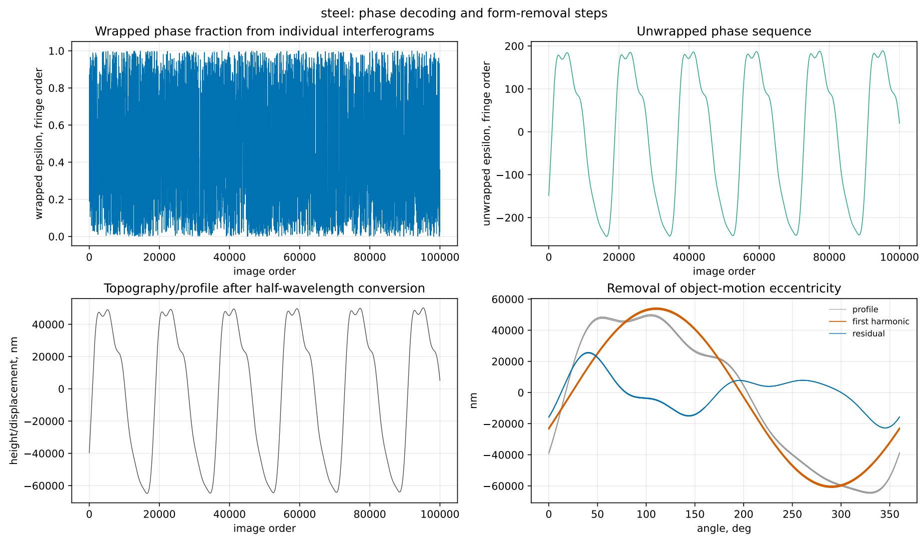

Figure 6: Steel sequence processing. (Top left) Wrapped excess fraction. (Top right) Unwrapped fringe order with linear drift. (Bottom left) Nanometre height profile after detrending. (Bottom right) Gaussian low-pass filtered residual.

2. Supplementary Validation Analyses

2.1. Descriptor comparison and uncertainty

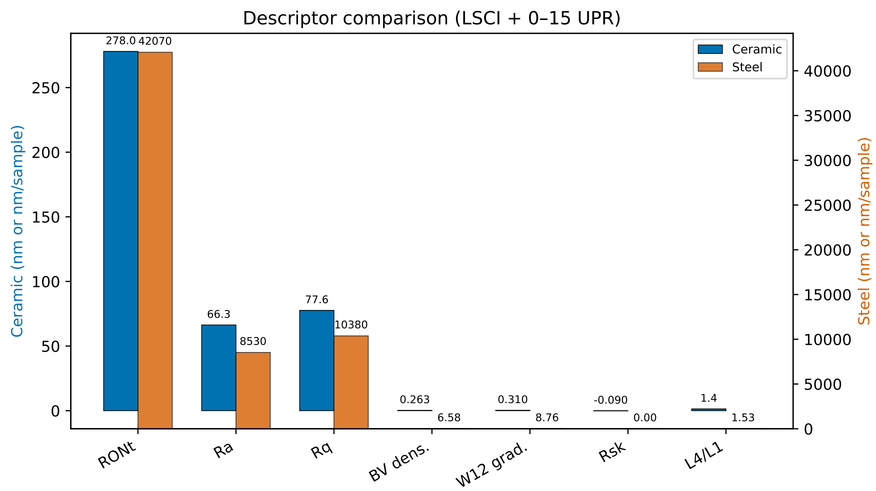

Figure 7: Comparison of ISO and gradient-based descriptors for both artefacts (bar charts). FN 111 values in blue, FN 101 values in orange. Error bars: 95% expanded within-run repeatability ($n = 5$).

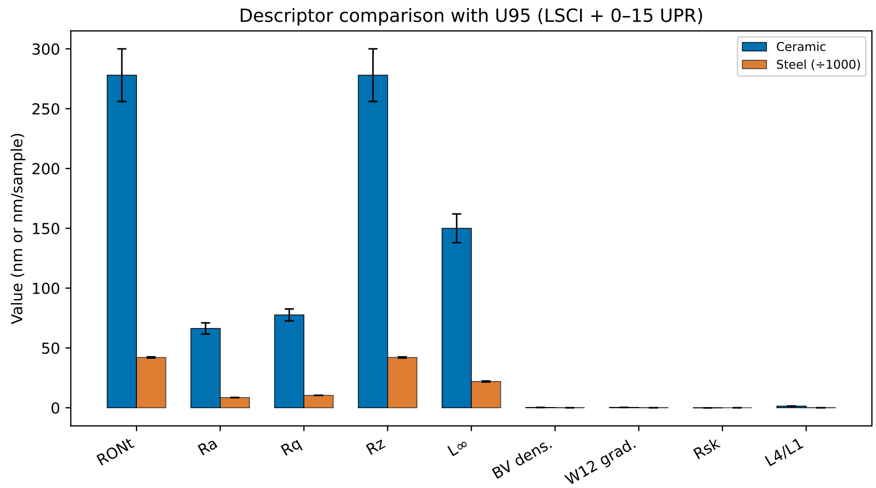

Figure 8: Uncertainty comparison: relative expanded uncertainty for each descriptor. Blue: FN 111; orange: FN 101.

2.2. *Descriptor independence validation*

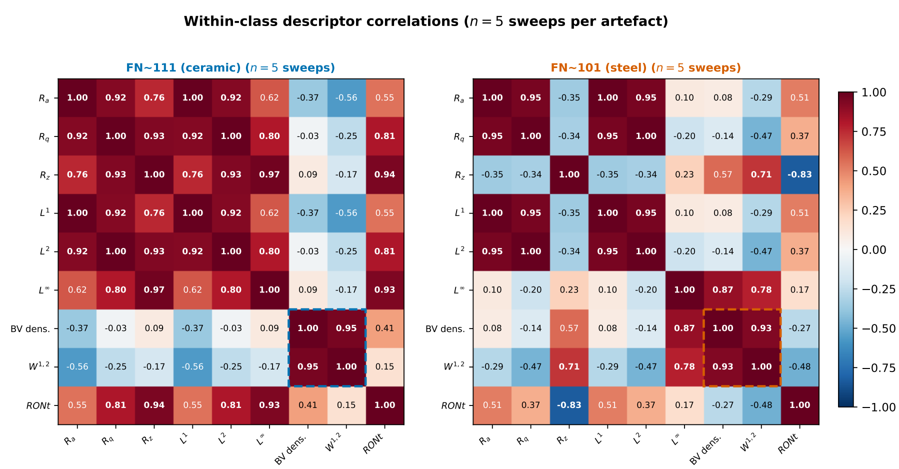

Figure 9: Within-class descriptor correlation matrices for FN 111 (left) and FN 101 (right). Dashed boxes highlight the Banach gradient-structure block.

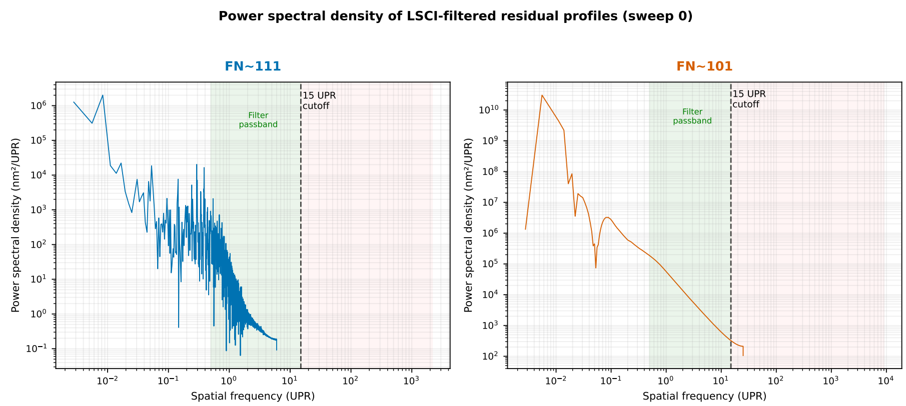

Figure 10: Power spectral density of LSCI-filtered residual profiles (sweep 0). Green shading: Gaussian filter passband (0.5–15 UPR).

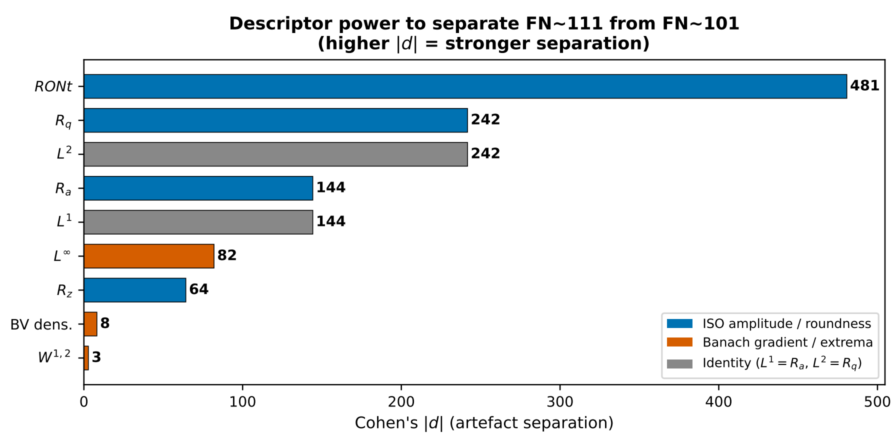

Figure 11: Cohen's $|d|$ effect size for separating FN 111 from FN 101. Blue: ISO amplitude/roundness. Orange: Banach gradient/extremum.

2.3. Critical descriptor assessment

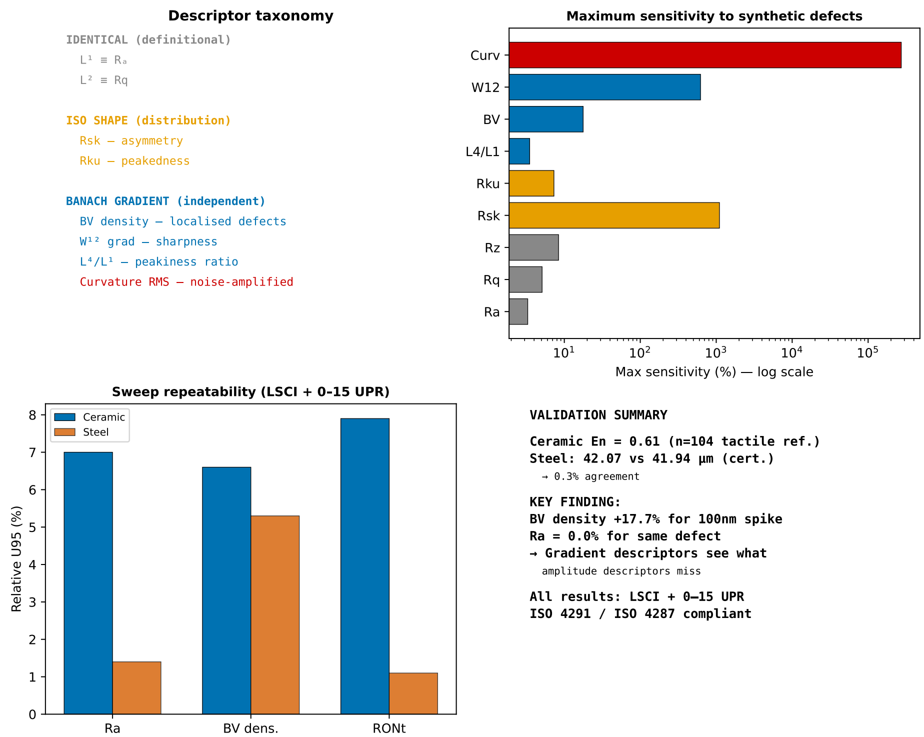

Figure 12: Critical assessment of gradient-based versus ISO descriptors. (Top left) Descriptor taxonomy. (Top right) Maximum sensitivity to synthetic defects. (Bottom left) Sweep-to-sweep stability. (Bottom right) Information-content summary.

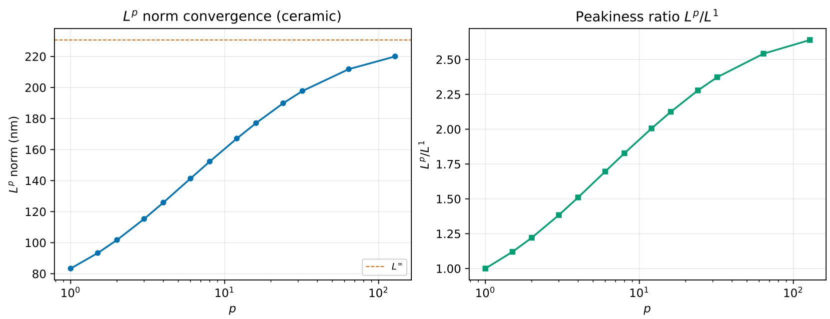

Figure 13: L^p norm convergence for the FN 111 residual profile (unfiltered). (Left) L^p as a function of p. (Right) Peakiness ratio L^p/L^1.

3. Supplementary Sensitivity and Artefact-Specific Analyses

3.1. Descriptor sensitivity to synthetic defects

Table 2: Sensitivity of ISO and gradient-based descriptors to synthetic profile defects. Values are percentage change from the unperturbed ceramic baseline (LSCI + Gaussian low-pass, 50% at 15 UPR; $R_a = 61.9$ nm, $R_q = 71.0$ nm). Bold entries indicate changes exceeding 10%. Defect details: Spike = single 100 nm point; Scratch = 5×50 nm depression; Waviness = 20 nm sinusoid ($\lambda = 500$ samples); Step = 30 nm offset; Noise = 5 nm RMS white noise.

Descriptor	Spike	Scratch	Waviness	Step	Noise
		Amplitude-averaging ISO parameters			
R_a	−0.0%	+0.1%	+3.2%	+3.3%	+0.1%
R_q	−0.0%	+0.1%	+3.3%	+5.1%	+0.2%
R_z (PV)	+0.0%	+0.0%	+7.0%	−2.6%	+8.4%
		ISO shape parameters			
R_{sk} (skewness)	−0.4%	+6.7%	**−168.5%**	**−1103.7%**	−0.1%
R_{ku} (kurtosis)	+0.0%	+0.1%	+2.4%	+7.3%	+0.4%
		Gradient-structure descriptors			
L^4/L^1 peakiness	+0.0%	+0.0%	+0.7%	+3.5%	+0.2%
BV density	**+17.7%**	+8.9%	**+11.7%**	+2.7%	**+2077.4%**
$W^{1,2}$ grad. seminorm	**+621.8%**	**+271.2%**	**+16.9%**	**+81.7%**	**+2260.9%**
Curvature RMS	**+273 161%**	**+111 458%**	**+94%**	**+47 230%**	**+901 073%**

The results lead to five conclusions:

1. Amplitude-averaging ISO descriptors (R_a, R_q, R_z) show negligible sensitivity to localised defects. A single 100 nm spike changes R_a by 0.0%, R_q by 0.0%, and R_z by 0.0%.
2. ISO shape parameters respond to asymmetry but not to symmetric localised features. R_{sk} changes by -1104% for a step defect (strong asymmetry), but only -0.4% for a symmetric spike. R_{ku} is insensitive to all tested defects.
3. Gradient-structure descriptors provide substantially higher sensitivity to localised defects than any single ISO 4287 parameter. BV density increases by $+18\%$ for a spike, $+9\%$ for a scratch, and $+12\%$ for waviness. The $W^{1,2}$ gradient seminorm shows even larger relative changes ($+622\%$ for spike, $+271\%$ for scratch).
4. R_{sk} and BV density are complementary: R_{sk} detects asymmetry, BV density detects localised gradients regardless of symmetry. Their combination provides descriptive sensitivity beyond any single classical parameter.
5. Curvature RMS is extremely sensitive to localised defects but strongly sampling-dependent; it should be used only for same-instrument comparisons.

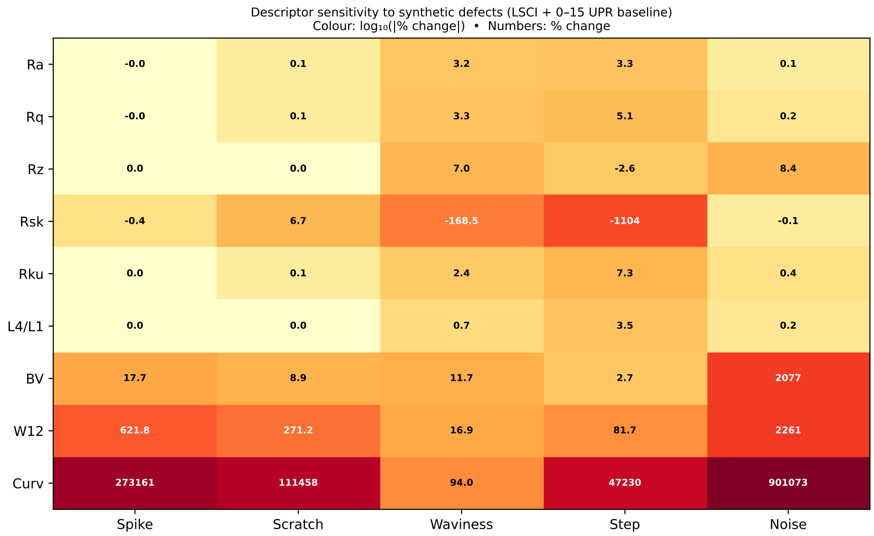

Figure 14: Sensitivity of ISO and gradient-based descriptors to five synthetic defect types injected into the FN 111 residual profile. Values are percentage change from the unperturbed baseline. Bold labels indicate changes exceeding 10%.

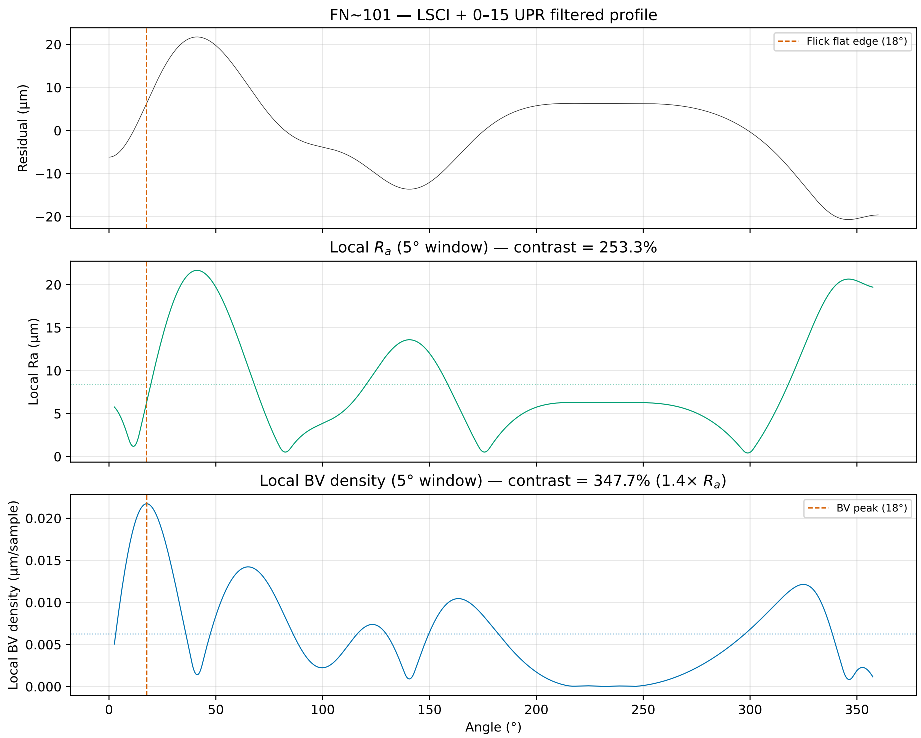

Figure 15: Flick flat detection on the JENOPTIK magnification standard FN 101. BV density (bottom panel) shows a clear spike at the Flick flat angular position, whereas the ISO 4287 amplitude parameter R_a (computed in a sliding window, top panel) shows negligible response. This demonstrates the complementary gradient-sensitive information provided by BV density beyond amplitude averaging.

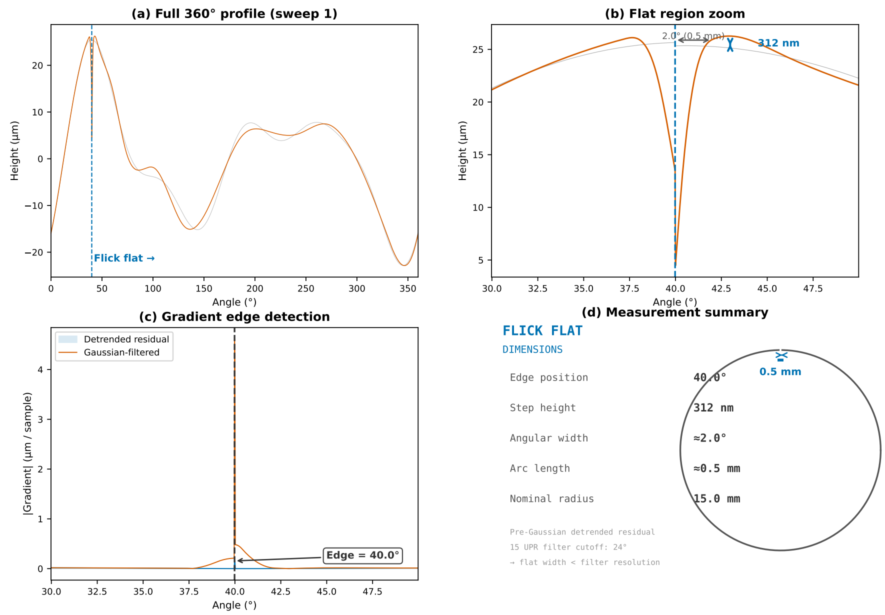

Figure 16: Flick flat measurement on the JENOPTIK magnification standard FN 101. Reconstructed profile across the Flick flat sector showing the characteristic step-like geometry, with the flat region and adjacent roundness profile labelled. $RONt = 42.07$ μm is dominated by the Flick flat depth.

3.3. Cross-instrument validation: optical versus tactile

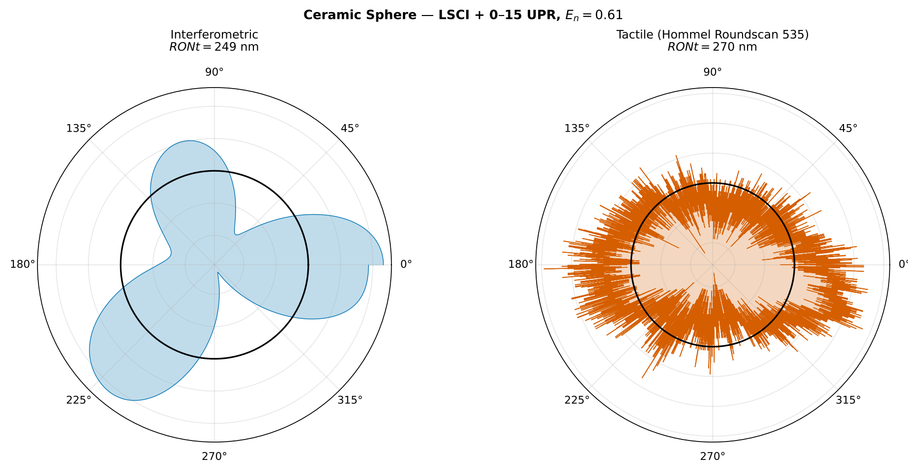

Figure 17: Polar comparison of optical (this work, blue) and tactile reference (Hommel Roundscan 535, orange) roundness profiles for the FN 111 standard. Both profiles are LSCI-centred and Gaussian low-pass filtered (50% at 15 UPR). The optical profile agrees with the tactile reference to within the combined uncertainty ($E_n = 0.61$).

4. Supplementary Artefact Documentation

4.1. Artefact photographs

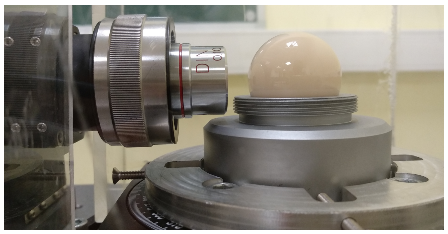

Figure 18: JENOPTIK roundness standard FN 111 (Al_2O_3, $\phi = 29.9588$ mm). Test report Art. 521 799. DAkkS-DKD calibration certificate Art. 532 529.

Figure 19: JENOPTIK magnification standard FN 101. Test report Art. 521 809. DAkkS-DKD calibration certificate Art. 532 528.

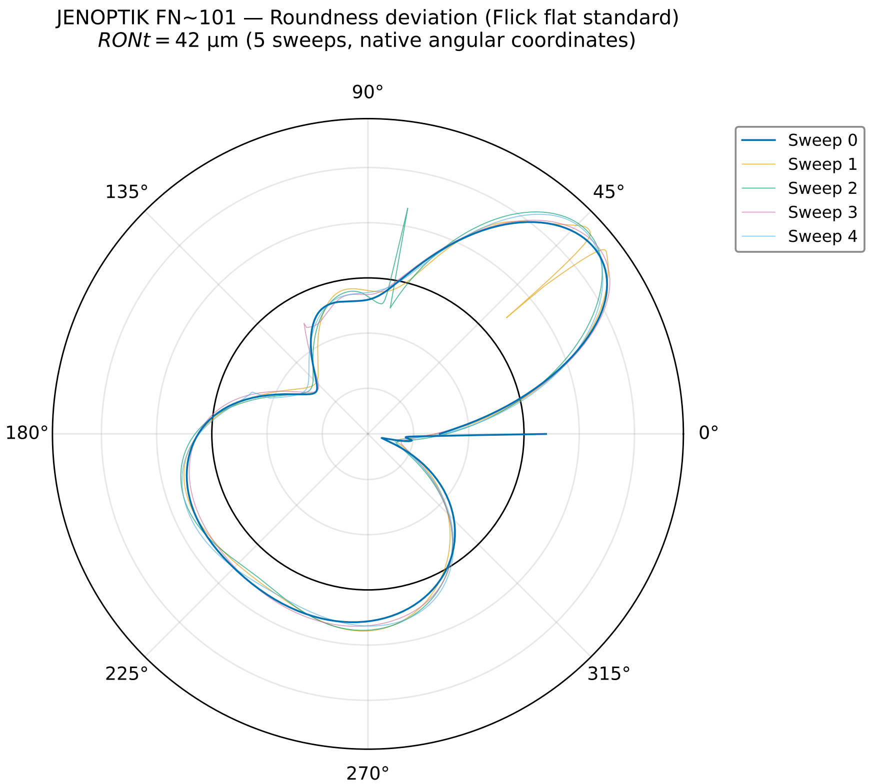

Figure 20: Steel roundness in polar coordinates (LSCI + Gaussian low-pass, 50% at 15 UPR). Five sweeps at native angular positions. The Flick flat appears at a shifted angular position in each sweep due to the 40° offset between successive 360° segments. $RONt = 42.07 \pm 0.48\ \mu$m.

4.3. Profile reconstruction plots

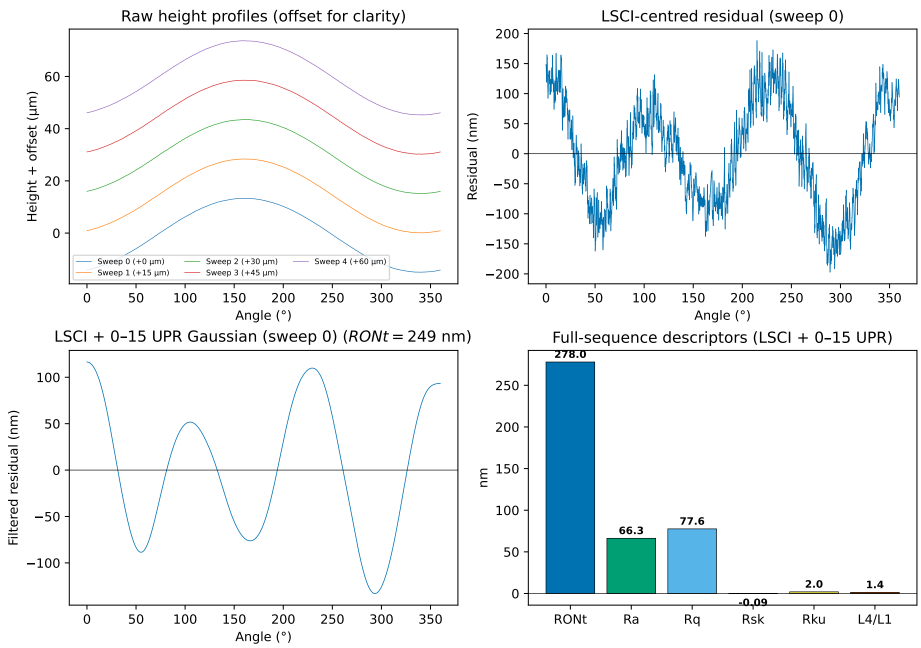

Figure 21: Ceramic profile reconstruction. (Top left) Raw height profiles from all five sweeps, vertically offset for clarity. (Top right) LSCI-centred residual for sweep 0. (Bottom left) Gaussian low-pass filtered residual (50% at 15 UPR) with $RONt = 278$ nm. (Bottom right) Full-sequence ISO and gradient-based descriptors.

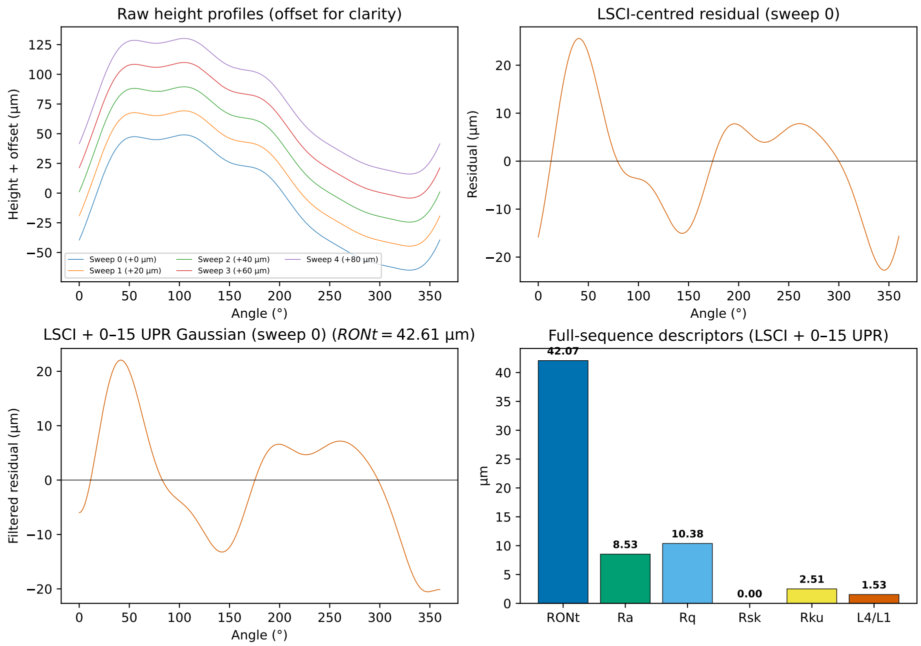

Figure 22: Steel profile reconstruction. (Top left) Detrended height profiles from all five sweeps, vertically offset. (Top right) LSCI-centred residual for sweep 0. (Bottom left) Gaussian low-pass filtered residual (50% at 15 UPR) with $RONt = 42.07\ \mu$m. (Bottom right) Full-sequence ISO and gradient-based descriptors.